

Using AWS KMS

(LAB-M13-01)

Overview

In this lab, you will:

- Configure AWS KMS to create your own customer master key (CMK)
- Encrypt EBS volumes and S3 data using AWS KMS
- Verify the behavior of S3 cross-region replication on server-side encrypted objects

Objectives

After completing this lab, you will be able to:

- Configure AWS KMS
- Encrypt Amazon EBS volumes using AWS KMS
- Encrypt Amazon S3 data using AWS KMS

Task 1: Create EC2 Instance

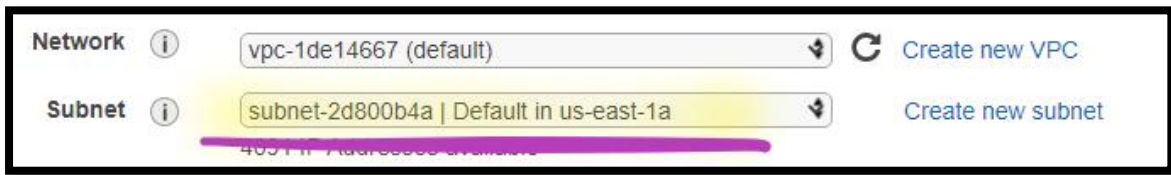
In this task, you will deploy EC2 instance and create file and folders in the Root volume.

Step 1: Create Amazon EC2 Instance

1. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
2. Dropdown and select **US East (N. Virginia)** region.
3. Click **Launch Instance**.
 - a. In the **Choose an Amazon Machine Image (AMI)** page:
 - i. Select the **Microsoft Windows Server 2019 Base**.
 - ii. Click **Select**.
 - b. In the **Choose an Instance Type** page:
 - i. Select the **t2.micro**.
 - ii. Click **Next: Configure Instance Details**.

c. In the **Configure Instance Details** page:

- i. **Subnet:** Dropdown and select the **First Availability Zone** (1a).



- ii. Select **Next: Add Storage**.

d. In the **Add Storage** page:

Note: Leave all the settings as default.

- i. Select **Next: Add Tags**.

e. In the **Add Tags** page:

- i. Click on **Add Tag**:
 - a) **Key:** Write **Name**.
 - b) **Value:** Write **Windows Server**.
- ii. Select **Next: Configure Security Group**.

f. In the **Configure Security Group** page:

- i. **Assign a security group:** Select **Create a new security group**.
 - a) **Security group name:** Write **WindowsServer-SG**.
 - b) **Description:** Write **Windows Server Security Group**.

Info: **RDP Port [3389]**, already added and allowed from anywhere [0.0.0.0/0].

- ii. Select **Review and Launch**.

g. In the **Review Instance Launch** page:

- i. Select **Launch**.

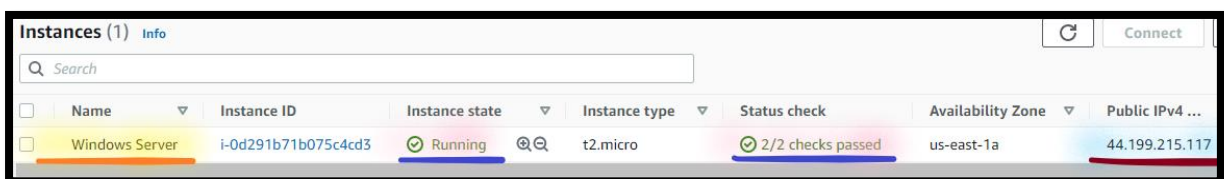
- a) In the **Select an existing key pair or create a new key pair** section:
- 1) **First selection box:** Dropdown and select **Choose an existing key pair**.
 - 2) **Key pair name:** Dropdown and select **My-SEC-LAB-KP**.
 - 3) **Enable** the **Checkmark** against the **I acknowledge that I have access ...**

Note: Leave the other settings as default.

- ii. Click on **Launch Instances**.
- h. Select the **View Instances**.

Step 2: Check the Windows Server Status

4. **From** the **EC2** console, click **Instance**.
5. Select **Windows Server**.
 - a. **Wait** for the **Instance State** to change to **Running** state.
 - b. **Wait** for the **Status check** to change to **2/2 checks passed**.
 - c. **Copy** the **Public IP address** in **Notepad**.



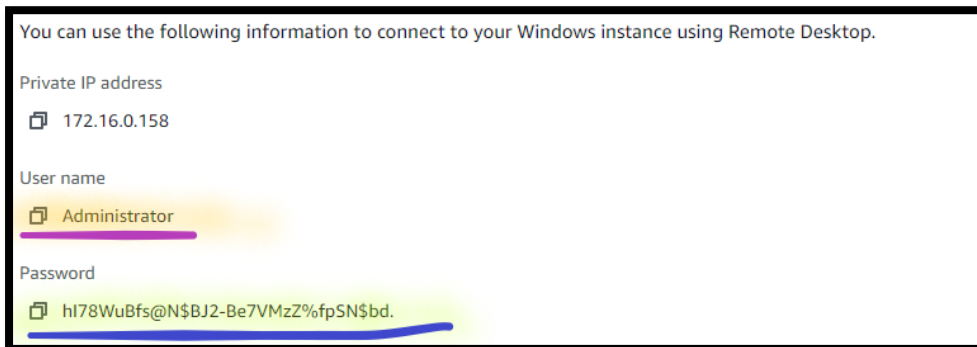
Instances (1) Info							
	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 ...
☐	Windows Server	i-0d291b71b075c4cd3	Running	t2.micro	2/2 checks passed	us-east-1a	44.199.215.117

Step 3: Generate Windows Password

6. **From** the **EC2** console, click **Instance**.
7. Select **Windows Server**.
 - a. Select **Actions**.
 - b. Select **Security**.

- c. Select **Get Windows Password**.
 - i. **Browse**: Navigate and select **My-SES-LAB-KP.pem** key pair.
 - ii. Click on **Decrypt Password**.

Note: Windows will pop-up with **user name** and **password**.



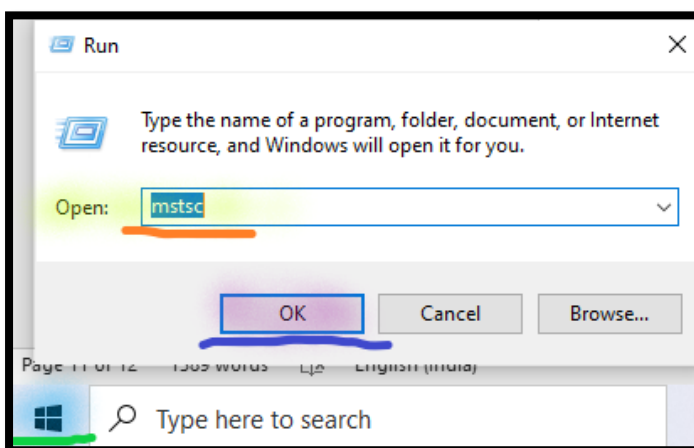
Note: Copy the **user name** and **password** in Notepad.

- iii. Select **Close**.

Step 4: Create File and Folders in Root Volume

In this task, you will create file and folders in the Root volume.

8. **From** the **local Desktop/ Laptop**, right click on **Start** & **Run**.
 - a. In the open, write **mstsc**, press **Ok**.



- b. **Type** the **Public IP Address** of the **Windows Server** instance.

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- c. Click **Connect**.
- d. **Type** the **Username** and **Password** of the **Windows Server** instance.
 - i. Click **Ok**.
 - ii. Click on **Yes** to confirm this connection, once ***prompted with the security message***.

Create File and Folders

- 9. **From** the **Windows Server**:
 - a. Go to **Start** menu, right click on **Start** & **Run**.
 - i. In the open, write **Explorer.exe**.
 - ii. Click on **This PC**.
 - b. Create **Root** folder in **C drive**.
 - c. Create **few Text** files in **Root** folder.

Note: Go to next task, But **don't logout** from **Windows server**.

Task 2: Create AWS KMS

In this task, you will create and manage an AWS KMS customer master key (CMK).

Step 1: Create AWS KMS Key

- 10. In the **AWS Management Console**, on the **Services** menu, select **KMS**.
- 11. Dropdown and select **US East (N. Virginia)** region.
- 12. Select **Create a key**.
 - a. In the **Configure Key** section:
 - i. **Key type:** Select **Symmetric**.
 - ii. **Expand Advanced options:**

- a) **Key material origin:** Select **KMS**.
- b) **Regionality:** Select **Single-region key**.

Key type [Help me choose](#)

☒ **Symmetric**
A single encryption key that is used for both encrypt and decrypt operations

☐ **Asymmetric**
A public and private key pair that can be used for encrypt/decrypt or sign/verify operations

▼ **Advanced options**

Key material origin
[Help me choose](#)

☒ **KMS**

☐ External

☐ Custom key store (CloudHSM)

Regionality
You cannot change this setting after the key is created. [Help me choose](#)

☒ **Single-region key**
Never allow this key to be replicated into other regions

☐ Multi-region key
Allow this key to be replicated into other regions

- iii. Select **Next**.

b. In the **Add labels** section:

- i. **Alias:** Type **PCI-Data-Key**.
- ii. **Description:** Type **KMS key used for encrypting PCI data**.

Alias
You can change the alias at any time. [Learn more](#)

Alias

PCI-Data-Key

Description - optional
You can change the description at any time.

Description - optional

KMS key used for encrypting PCI data

iii. Click **Next**.

c. In the **Define key administrative permissions** section:

Note: Leave all the settings as default.

i. Click **Next**.

d. In the **Define key usage permissions** section:

Note: Leave all the settings as default.

i. Click **Next**.

e. In the **Review** section:

Note: Review all the details.

i. Click **Finish**.

Note: Wait, till KMS key gets created.

Step 2: Rotate AWS KMS Key

13. From the **AWS KMS Key** page.
14. Select **Customer managed keys**.
 - a. Open the **PCI-Data-Key**.

☒	Aliases ▾	Key ID ▾	Status	Key spec ⓘ	Key usage
☒	PCI-Data-Key	e173b87f-173a-43b0-a4c2-afcbc2fe7eaf	Enabled	SYMMETRIC_DEFAULT	Encrypt and decrypt

- b. Select the **Key rotation**.
 - i. **Enable** the **Checkmark** against **Automatically rotate this CMK every year**.

Key policy	Cryptographic configuration	Tags	Key rotation	Aliases
Key rotation				
☒ Automatically rotate this KMS key every year. Learn more				

- ii. Click **Save**.

Task 3: Encrypt Amazon EBS Volume using AWS KMS

For compliance reasons, it might be important to encrypt the storage disks of your Amazon EC2 instances. In this task, you will see how to use AWS KMS to encrypt an Amazon EBS volume and how to use AWS KMS to encrypt the boot volume of a server.

Step 1: Create Encrypted Amazon EBS volume

In this task, you will use your CMK to encrypt EBS volumes.

15. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
16. Click **Volumes**.
17. Select **Create volume**, and configure:
 - a. **Volume Type**: Dropdown and select **General Purpose SSD (gp2)**.

- b. **Size (GiB):** Write **1**.
- c. **Availability Zone:** Choose the **First Availability Zone** (1a).

Volume type [Info](#)
General Purpose SSD (gp2)

Size (GiB) [Info](#)
1
Min: 1 GiB, Max: 16384 GiB. The value must be an integer.

IOPS [Info](#)
100 / 3000
Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) [Info](#)
Not applicable

Availability Zone [Info](#)
us-east-1a

- d. **Encryption:** **Enable** the **Checkmark** against **Encrypt this volume**.
 - i. **KMS key:** Dropdown and select **PCI-Data-Key**.

Encryption [Info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☒ Encrypt this volume

KMS key [Info](#)
PCI-Data-Key

Info: Encryption can't be after volume created

- e. **Tag:** Select **Add tag** and **configure:**
 - i. **Key:** Write **Name**.
 - ii. **Value:** Write **WindowsServer-DataDisk**.
- f. Select **Create Volume**.

Note: **Wait**, till the Volume status is showing as **available**.

Note: You will see, **Boot volume** as **Not Encrypted** and **Data Disk** as **Encrypted**.

Volumes (2)									
	Filter volumes								
☐	Name	Volu...	Size	Availa...	Volume state	Volu...	Encryption	KMS key ID	KMS key alias
☐	Windows Server	vol-0826d...	30 GiB	us-east-1a	In-use	Okay	Not encrypted	-	-
☐	WindowsServer-DataDisk	vol-0d644...	1 GiB	us-east-1a	Available	Okay	Encrypted	e173b87f-173...	PCI-Data-Key

Step 2: Attach an Encrypted Amazon EBS Volume to an Instance

18. From the **EC2** console, click **Volumes**.

19. Select **WindowsServer-DataDisk** volume.

Note: Volume status is showing as **available**.

- a. Choose **Actions**.
- b. Select **Attach Volume**.
- c. **Instance:** Click in the box.
 - i. Select **Windows Server** instance.

Volume ID

 vol-0d644b7a685aca937 (WindowsServer-DataDisk)

Availability Zone

us-east-1a

Instance [Info](#)

i-0d291b71b075c4cd3 

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

xvdf

Windows device names: xvdf through xvdp

- ii. Select **Attach volume**.

Step 3: Mount EBS Volume

In this task, you will use your encrypted EBS volume to attach to EC2 instance.

20. **Login** into **Windows Server** instance via **RDP** using **Public IP address**.

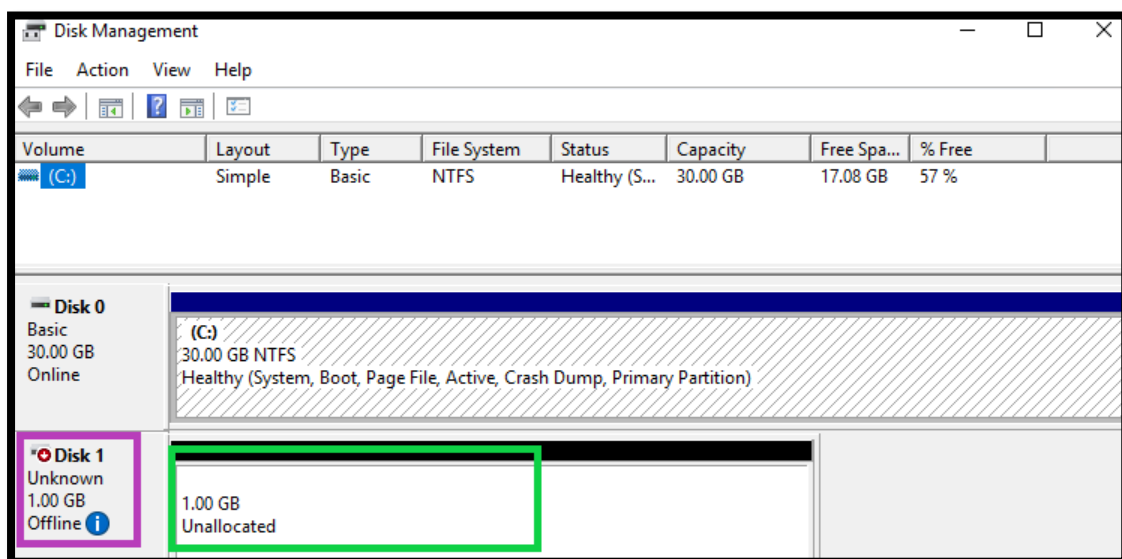
Note: Login using the **Administrator** and **Windows Password** (which you have generated in the previous step).

21. **From** the **Windows Server**:

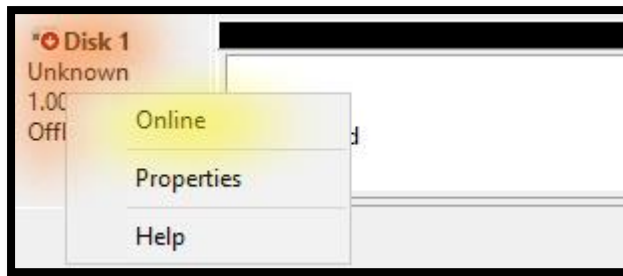
- Go to **Start** menu, right click on **Start** & **Run**. In the open, write **diskmgmt.msc**.
- The new disk appears as **Unallocated**.

Note: If you are unable to see the disk, **Scroll down in Disk Management tool** to view the disk.

Note: Your new volume shown as **Offline**.

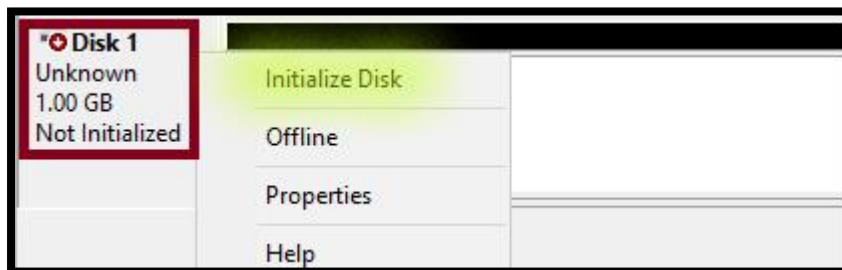


- Right-click** on the disk (**within purple portion**) and select **Online**.

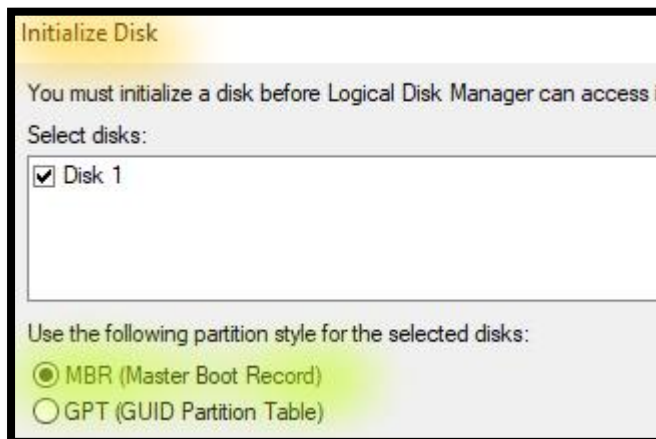


Note: Your new volume shown as **Not Initialised**.

- d. **Right-click** on the disk (**within brown portion**) and select **Initialise Disk**.



- e. **During initialise**, if it prompts you to select the partition Style, Select **MBR (master Boot Recorder)**, select **Ok**



Note: Your new volume shown as **Online**.

- f. **Right-click** on the disk (**within black portion**) and select **New simple volume**.

- i. In the **Welcome to the New Simple Volume Wizard** page:

Note: Leave the other settings as default.

- a) Select **Next**.
- ii. In the **Specify Volume Size** page:

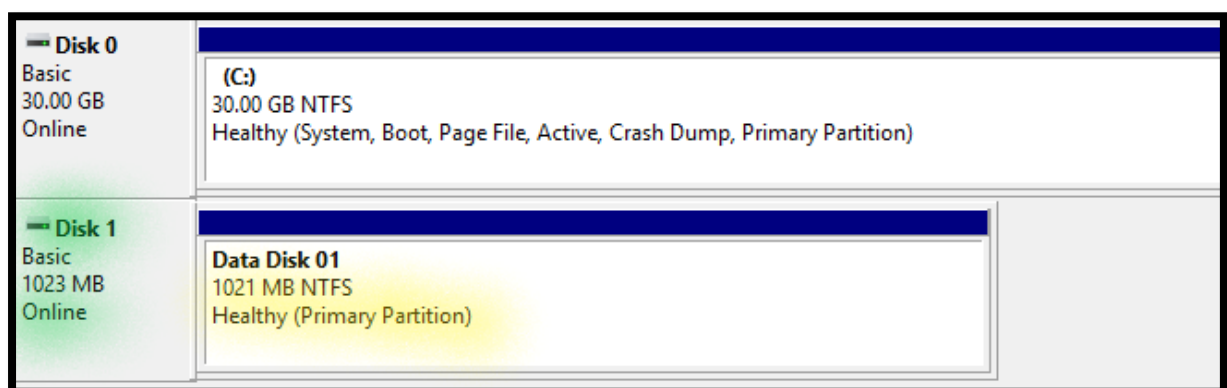
Note: Leave the other settings as default.

- a) Select **Next**.
- iii. In the **Assign Drive Letter or Path** page:

Note: Leave the other settings as default.

- a) Select **Next**.
- iv. In the **Format Partition** page:
 - a) **Volume Label:** Write **DataDisk**.
 - b) Select **Next**.
- v. In the **Completing the Simple Volume Wizard** page:
 - a) Select **Finish**.

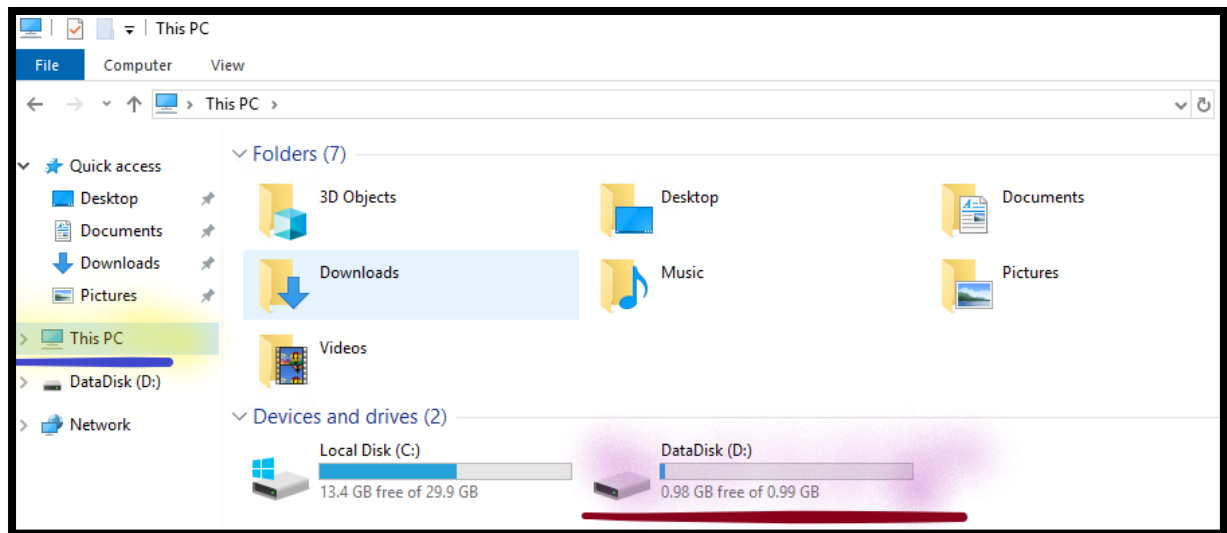
Note: Once completed disk will be showing as **Healthy**. **Close disk management tool**.



Create File and Folders

22. From the **Windows Server**:

- a. Go to **Start** menu, right click on **Start** & **Run**.
 - i. In the open, write **Explorer.exe**.
 - ii. Click on **This PC**.



- b. Create **KMS-Disk** folder in **D drive**.
- c. Create **few Text** files in **KMS-Disk** folder.

Note: Go to next task, But **don't logout** from **Windows server**.

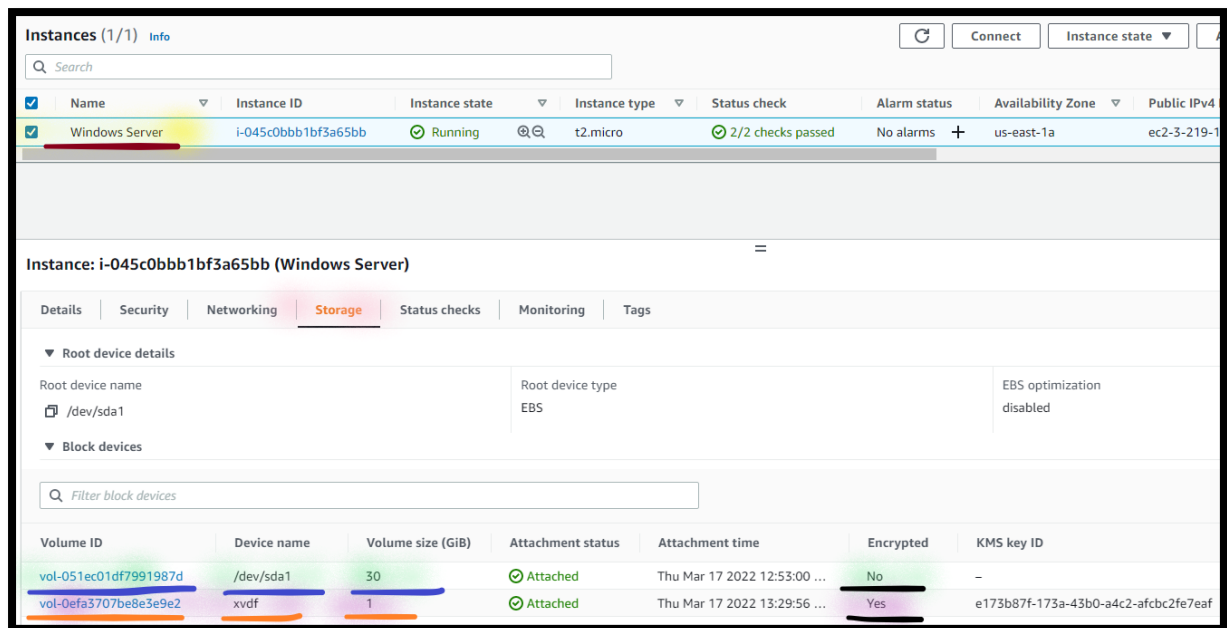
Step 4: Check the Windows Server Storage Status

23. From the **EC2** console, click **Instance**.

24. Select **Windows Server**.

- a. Select **Storage**.

Note: You will see, **Boot volume** as **Not Encrypted** and **Data Disk volume** as **Encrypted**.



Task 3: Encrypt a Boot volume using AWS KMS

In this section, you will use your CMK to encrypt the boot volume of an Amazon EC2 instance. This can be important when data is stored on the boot volume and you need the data to be encrypted at rest.

To encrypt a boot volume, you first need to make an Amazon Machine Image (AMI) of the instance. Then, you will encrypt the AMI by copying it. When you launch an instance from the encrypted AMI, the instance boot volume will be encrypted.

Step 1: Create Customise Image

25. From the **EC2** console, click **Instance**.

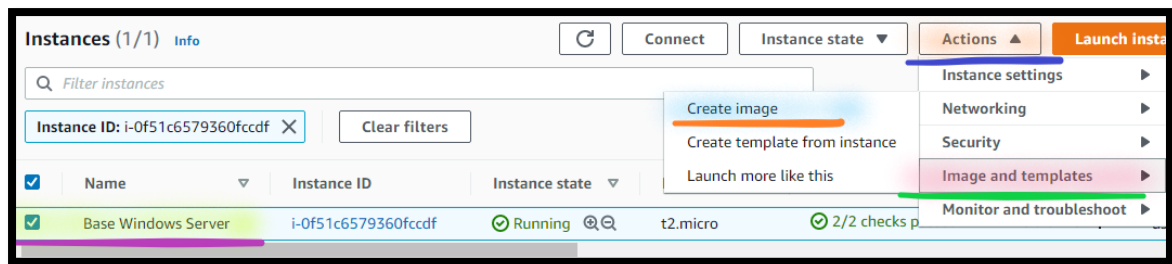
26. Select **Instances**.

a. Select **Windows Server**.

i. Select **Actions**.

ii. Select **Image and templates**.

iii. Select **Create Image**.



iv. In the **Create image** page:

- a) **Image name:** Write **WindowsServerImage-01**.
- b) **Image description:** Write **Image from Windows Server BootVolume-Unencrypted**.

Note: You will see , **Boot volume** as **Not Encrypted** and **Data Disk** as **Encrypted**.

Instance volumes								
Volume type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
EBS	xvdf	Create new snapshot fr...	1	EBS General Purpose SS...	100		☐ Enable	☒ Enable
EBS	/dev/s...	Create new snapshot fr...	30	EBS General Purpose SS...	100		☒ Enable	☐ Enable

iii. **Tag:** Select **Add tag** and **configure:**

- a) **Key:** Write **Name**.
- b) **Value:** Write **WindowsServerImage-01**.

v. Click on **Create Image**.

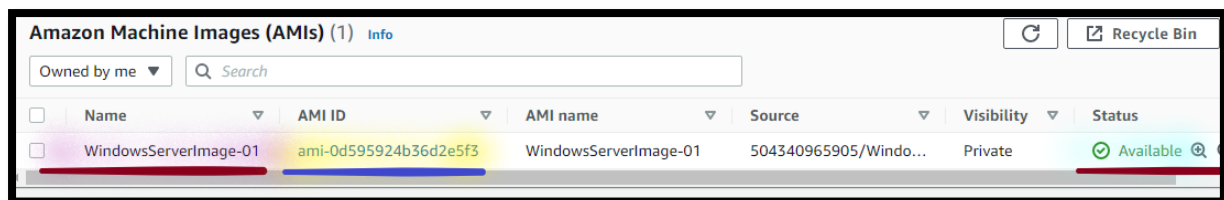
Step 2: Check the AMI and Snapshot Status

27. From the **EC2** console, click **AMIs**.

Info: By default, create-image will restart the current instance before creating the AMI, in order to ensure the integrity of the image on the file system.

Note: You will see image **Status** as **Pending**.

Note: **Wait** (~ 5 minutes) till **Status** changed to **Available**.



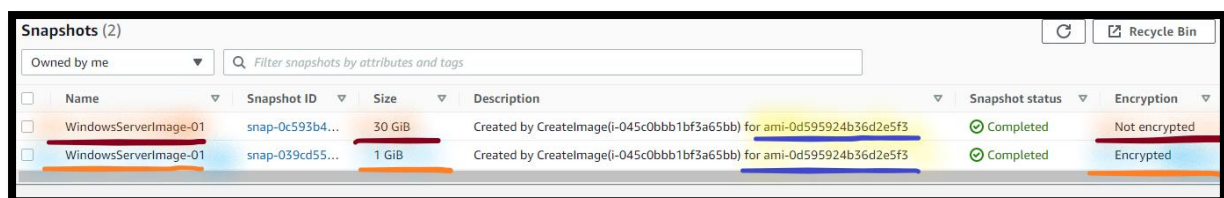
Amazon Machine Images (AMIs) (1) Info

Owned by me Search

	Name	AMI ID	AMI name	Source	Visibility	Status
☐	WindowsServerImage-01	ami-0d595924b36d2e5f3	WindowsServerImage-01	504340965905/Windo...	Private	Available

28. From the **EC2** console, click **Snapshots**.

Note: You will see, **Boot volume Snapshot** as **Not Encrypted** and **Data Disk Snapshot** as **Encrypted**.



Snapshots (2)

Owned by me Filter snapshots by attributes and tags

	Name	Snapshot ID	Size	Description	Snapshot status	Encryption
☐	WindowsServerImage-01	snap-0c593b4...	30 GiB	Created by CreateImage(i-045c0bbb1bf3a65bb) for ami-0d595924b36d2e5f3	Completed	Not encrypted
☐	WindowsServerImage-01	snap-039cd55...	1 GiB	Created by CreateImage(i-045c0bbb1bf3a65bb) for ami-0d595924b36d2e5f3	Completed	Encrypted

Step 3: Create Second AWS KMS Key

29. In the **AWS Management Console**, on the **Services** menu, select **KMS**.

30. Dropdown and select **US East (N. Virginia)** region.

31. Select **Create a key**.

a. In the **Configure Key** section:

i. **Key type:** Select **Symmetric**.

ii. **Expand Advanced options:**

a) **Key material origin:** Select **KMS**.

b) **Regionality:** Select **Multi-region key**.



Regionality

You cannot change this setting after the key is created. [Help me choose](#)

☐ Single-region key
Never allow this key to be replicated into other regions

☒ Multi-region key
Allow this key to be replicated into other regions

Info: AWS KMS supports multi-Region keys, which are AWS KMS keys in different AWS Regions. In multiple regions, each set of related multi-Region keys has the same key material and key ID, so you can encrypt data in one AWS Region and decrypt it in a different AWS Region without re-encrypting or making a cross-Region call to AWS KMS.

iii. Select **Next**.

b. In the **Add labels** section:

i. **Alias:** Type **PCI-Data-Key-02**.

ii. **Description:** Type **KMS key used for encrypting PCI data**.

iii. Click **Next**.

c. In the **Define key administrative permissions** section:

Note: Leave all the settings as default.

i. Click **Next**.

d. In the **Define key usage permissions** section:

Note: Leave all the settings as default.

- ii. Click **Next**.
- e. In the **Review** section:

Note: Review all the details.

- ii. Click **Finish**.

Note: Wait, till KMS key gets created.

Step 4: Copy AMI with new KMS Key

- 32. **From** the **EC2** console, click **AMIs**.
 - a. Select **WindowsServerImage-01**.
 - i. Select **Actions**.
 - ii. Select **Copy AMI**.
 - a) **AMI Copy Name:** Write **WindowsServerImage-02**.
 - b) **Description:** Write **Image from Windows Server BootVolume_DataDisk-Unencrypted**.
 - c) **Destination Region:** Dropdown and select **US East (N. Virginia)** region.
 - d) **Enable** the **Checkmark** against **Encrypt EBS snapshots of AMI copy**.
 - 1) **KMS key:** Dropdown and select **PCI-Data-Key-02**.

Original AMI ID
ami-008123d37aedc8e6d (WindowsServerImage-01)

AMI copy name
WindowsServerImage-02

AMI copy description
Image from Windows Server BootVolume_DataDisk-Unencrypted

Destination Region
A copy of the original AMI will be created in the destination Region.
N. Virginia (US East)

☒ Encrypt EBS snapshots of AMI copy
Encrypts all snapshots in the AMI copy with the same key.

KMS key
This is the KMS key used to encrypt the snapshots.
PCI-Data-Key-02

Info: Encryption keys used by an Amazon EBS volume can't be changed. However, you can Create a snapshot of the volume and then use the new encryption key.

Info: All the snapshots will be encrypted from the KMS Key, even though Volume encrypted with different KMS key.

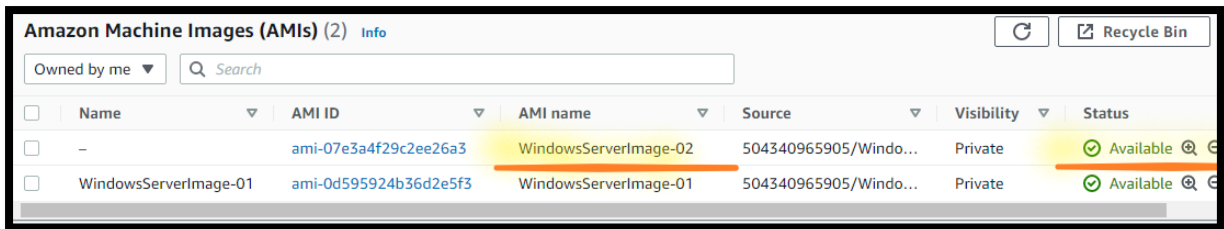
iii. Select **Copy AMI**.

Step 5: Check the AMI and Snapshot Status

33. From the **AMIs** page:

Note: You will see Image **Status** as **Pending**.

Note: **Wait** (~ 5 minutes) till **Status** changed to **Available**.



Amazon Machine Images (AMIs) (2) Info

Owned by me Search

	Name	AMI ID	AMI name	Source	Visibility	Status
☐	-	ami-07e3a4f29c2ee26a3	WindowsServerImage-02	504340965905/Windo...	Private	Available
☐	WindowsServerImage-01	ami-0d595924b36d2e5f3	WindowsServerImage-01	504340965905/Windo...	Private	Available

34. From the **EC2** console, click **Snapshots**.

Note: You will see, **Boot volume Snapshot** as **Encrypted** and **Data Disk Snapshot** as **Encrypted**.

Note: You will see, **Boot volume Snapshot** and **Data Disk Snapshot** **Encrypted** with **PCI-Data-Key-02**.



Snapshots (4)

Owned by me Filter snapshots by attributes and tags

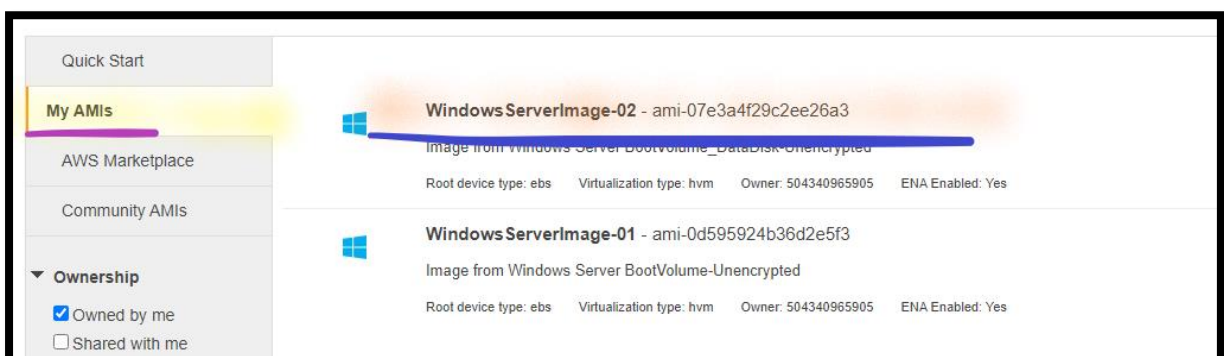
	Name	Snapshot ID	Size	Snapshot ...	Encryption	KMS key ID	KMS key alias
☐	WindowsServerImage-01	snap-027ab5aba08ef55d2	30 GiB	Completed	Not encrypted	-	-
☐	WindowsServerImage-01	snap-033a15353ac54f275	1 GiB	Completed	Encrypted	e173b87f-173a-43b0-a4c2-afbc2fe7eaf	PCI-Data-Key
☐	-	snap-0590e6f7a0427fdbb	30 GiB	Completed	Encrypted	6a46ee61-7be8-413b-840d-a4394620b179	PCI-Data-Key-02
☐	-	snap-01def631174f22d19	1 GiB	Completed	Encrypted	6a46ee61-7be8-413b-840d-a4394620b179	PCI-Data-Key-02

Step 6: Create Amazon EC2 Instance

35. From the **EC2** console, click **Instance**.

36. Click **Launch Instance**.

- a. In the **Choose an Amazon Machine Image (AMI)** page:
 - i. Select the **My AMIs**.
 - a) Select the **WindowsServerImage-02**.



b. In the **Choose an Instance Type** page:

- i. Select the **t2.micro**.
- ii. Click **Next: Configure Instance Details**.

c. In the **Configure Instance Details** page:

Note: Leave all the settings as default.

- i. Select **Next: Add Storage**.

d. In the **Add Storage** page:

Note: Leave all the settings as default.

- i. Select **Next: Add Tags**.

e. In the **Add Tags** page:

- i. Click on **Add Tag**:
 - a) **Key**: Write **Name**.
 - b) **Value**: Write **Windows Server-02**.
- ii. Select **Next: Configure Security Group**.

f. In the **Configure Security Group** page:

- i. **Assign a security group**: Click **Select an existing security group**.
 - a) Select the **WindowsServer-SG**.

g. Select **Review and Launch**.

i. In the **Review Instance Launch** page:

- a. Select **Launch**.

a) In the **Select an existing key pair or create a new key pair** section:

- 1) **First selection box**: Dropdown and select **Proceed without a key pair**.

- 2) **Enable** the **Checkmark** against the ***I acknowledge that I have access***

Note: Leave the other settings as default.

- ii. Click on **Launch Instances**.
- h. Select the **View Instances**.

Step 7: Check the Windows Server Status

- 37. **From** the **EC2** console, click **Instance**.
- 38. Select **Windows Server-02**.
 - d. **Wait** for the **Instance State** to change to **Running state**.
 - e. **Wait** for the **Status check** to change to **2/2 checks passed**.

Note: **Wait** for the server deployment.

Step 8: Check the Windows Server Storage Status

- 39. **From** the **EC2** console, click **Instance**.
- 40. Select **Windows Server-02**.
 - a. Select **Storage**.

Note: You will see, **Boot volume** as **Encrypted** and **Data Disk volume** as **Encrypted**.

Instances (1/2) Info

Search

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
Windows Server	i-045c0bbb1bf3a65bb	Running	t2.micro	2/2 checks passed	No alarms
Windows Server-02	i-0d2032aa659797d57	Running	t2.micro	-	No alarms

Instance: i-0d2032aa659797d57 (Windows Server-02)

Details | Security | Networking | **Storage** | Status checks | Monitoring | Tags

▼ Root device details

Root device name	Root device type
/dev/sda1	EBS

▼ Block devices

Filter block devices

Volume ID	Device name	Volume size (GiB)	Attachment status	Attachment time	Encrypted
vol-0cb738f523a17b9ef	/dev/sda1	30	Attached	Thu Mar 17 2022 16:39:47 ...	Yes
vol-0f241b7c436fe0797	xvdf	1	Attached	Thu Mar 17 2022 16:39:47 ...	Yes

Step 10: Verify File and Folders in Second Windows Server

In this task, you will verify the file and folders created in the Root and Data Disk volume in First server.

41. **From** the **local Desktop/ Laptop**, right click on **Start** & **Run**.
 - a. In the open, write **mstsc**, press **Ok**.
 - b. **Type** the **Public IP Address** of the **Windows Server-02** instance.
 - c. Click **Connect**.
 - d. **Type** the **Username** and **Password** of the **Windows Server** (which you have generated in the First server) instance.
 - i. Click **Ok**.
 - ii. Click on **Yes** to confirm this connection, once **prompted with the security message**.

42. **From the Windows Server-02:**

- a. Go to **Start** menu, right click on **Start** & **Run**.
 - i. In the open, write **Explorer.exe**.
 - ii. Click on **This PC**.
- b. **Open** the **Root** folder in **C drive**.
 - i. **Open** and **verify** the **Text** files in **Root** folder.
- c. **Open** the **KMS-Disk** folder in **C drive**.
 - i. **Open** and **verify** the **Text** files in **Root** folder.

Task 4: Encrypt Amazon S3 data using AWS KMS

In this task, you will encrypt Amazon S3 data using your CMK. This method of encryption is referred to as Amazon S3 server-side encryption with AWS KMS managed keys (SSE-KMS).

Step 1: Encrypt Amazon S3 data

43. In the **AWS Management Console**, on the **Services** menu, click **S3**.
44. Click **Create bucket**.

- a. In the **General configuration** section:
 - i. **Bucket name:** Write **secret-123**.

Note: Replace **123** with a random number to make bucket name unique.

- ii. **Region:** Dropdown and select **US East (N. Virginia)**.



The screenshot shows the AWS Management Console interface for creating a new S3 bucket. The 'General configuration' section is active. The 'Bucket name' field is populated with 'secret-123'. Below this field, a message states: 'Bucket name must be unique and must not contain spaces or uppercase letters. See rules for bucket naming'. The 'AWS Region' dropdown menu is open, showing 'US East (N. Virginia) us-east-1' as the selected option.

- b. In the **Object Ownership** section:
 - i. Select **ACL Enabled**.
 - ii. Select **Bucket owner preferred**.
- c. In the **Block Public Access settings for this bucket** section:
 - i. **Block all public access:** **Deselect** (*remove the checkmark*) the Block all public access option.
 - ii. **Select I acknowledge** that the current settings might result in this bucket and the objects within becoming public.

☐ **Block all public access**
 Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
 S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
 S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
 S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
 S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

 **Turning off block all public access might result in this bucket and the objects within becoming public**
 AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

- d. In the **Default encryption** section:
 - i. **Server-side encryption:** Select **Enable**.
 - ii. **Encryption key type:** Select **AWS Key Management Service key (SSE-KMS)**.
 - iii. **AWS KMS key:** Select **Choose from your AWS KMS keys**.
 - iv. **AWS KMS key:** Dropdown and select **PCI-Data-Key-02**.

Server-side encryption

☐ Disable

☒ Enable

Encryption key type

To upload an object with a customer-provided encryption key (SSE-C), use the AWS CLI, AWS SDK, or Amazon S3 REST API.

☐ Amazon S3-managed keys (SSE-S3)

An encryption key that Amazon S3 creates, manages, and uses for you. [Learn more](#)

☒ AWS Key Management Service key (SSE-KMS)

An encryption key protected by AWS Key Management Service (AWS KMS). [Learn more](#)

AWS KMS key

☐ AWS managed key (aws/s3)

arn:aws:kms:us-east-1:504340965905:alias/aws/s3

☒ Choose from your AWS KMS keys

☐ Enter AWS KMS key ARN

AWS KMS key

arn:aws:kms:us-east-1:504340965905:key/e173b... ▼

Refresh

Create key

Note: Leave other details as default.

- e. Click **Create bucket**.

Step 2: Upload the Object in the Bucket

45. From the **S3** console, click **Buckets**.

46. Open **secret-123** bucket.

- a. Click **Objects**.
- b. Click **Upload**.
- c. Select **Add files**.
- d. Navigate and select **mydata.txt** file.

Note: **mydata.txt** file available with the lab manual.

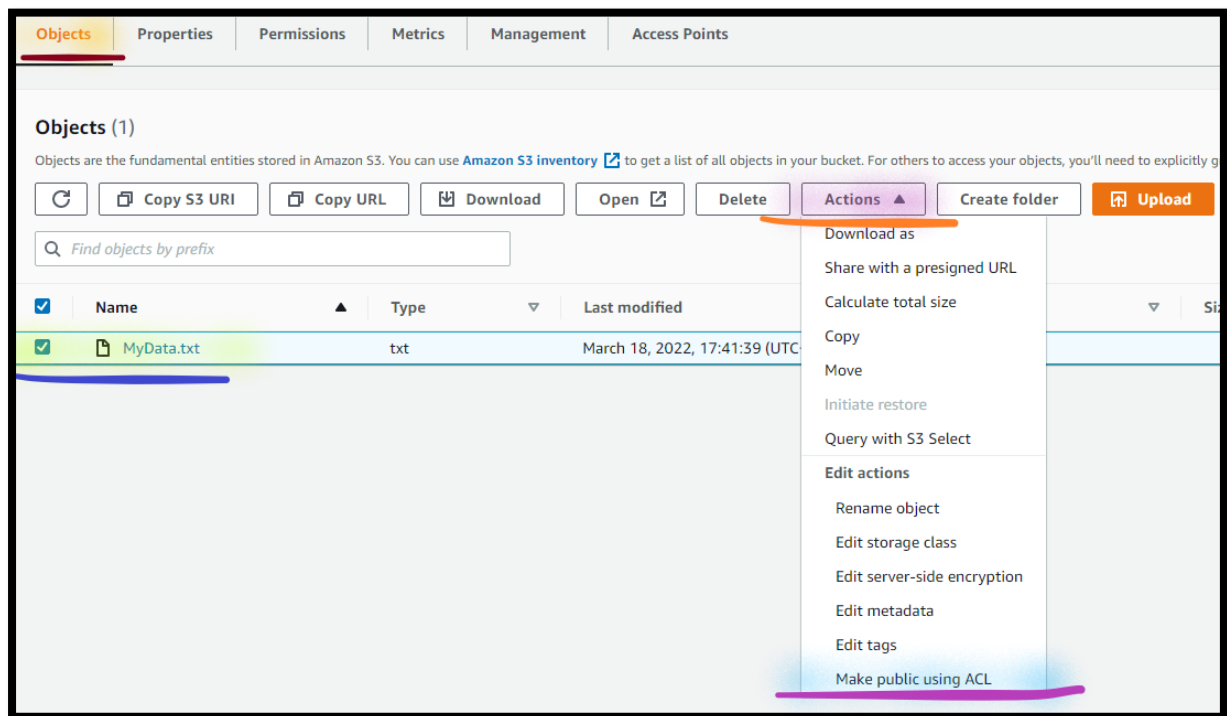
- e. Click **Upload**.

Note: **Wait** till all the contents uploaded successfully and can get the message **Upload succeeded**.

- f. Click **Close**.

Step 3: Make Object Public

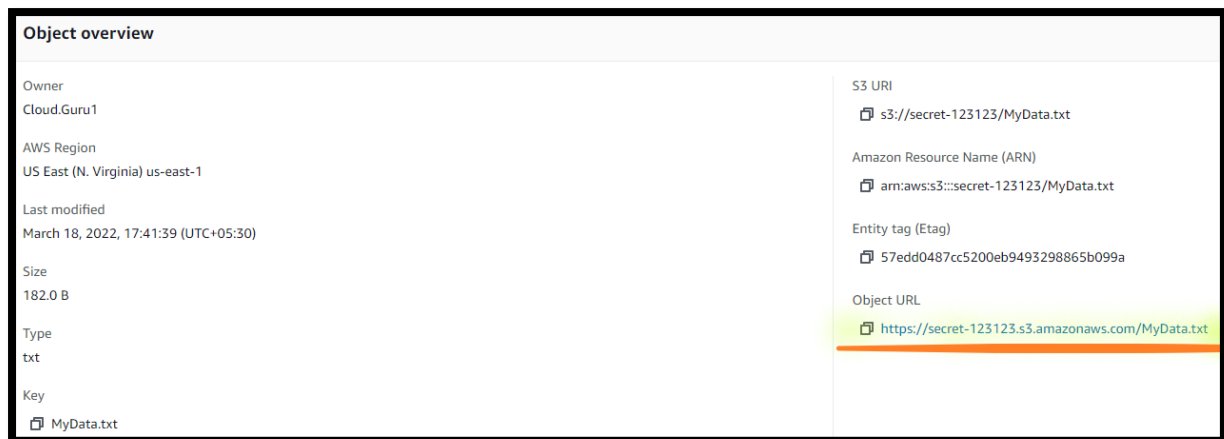
47. From the **S3** console, click **Buckets**.
48. Open **secret-123** bucket.
 - a. Select **mydata.txt**.
 - i. Select **Actions**.
 - ii. Select **Make public using ACL**.



- a) Select **Make public**.
- b) Select **Close**.

Step 4: Access the Public Object

49. From the **S3** console, click **Buckets**.
50. Open **secret-123** bucket.
 - a. Open **mydata.txt**.
 - b. Copy **Object URL** in the **Notepad**.



17. **From** your **local Desktop/ Laptop** open the **Browser**, type **Object URL** of **MyData.txt** and access your **Object**.

Note: You are **not able to open the file**.

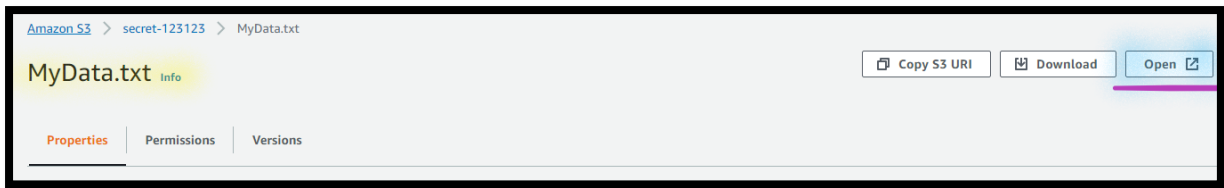
Info: Notice that you are not able to open the file, even though it is public, because the file is encrypted and needs to be decrypted before you can view it. The error message also tells you that the reason the file cannot be opened is because the request requires an AWS KMS key managed by AWS.



18. **From** the **S3** console, click **Buckets**.

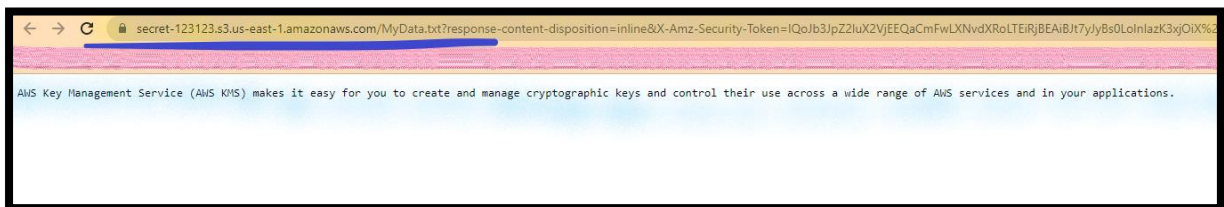
19. Open **secret-123** bucket.

- Open **mydata.txt**.
- Click the **Open**.



Note: You are **able to open the file**.

Info: You are able to see the file because Amazon S3 transparently decrypts the file using your CMK as long as the request comes from a source that has permission to use the CMK.



Note: You can also use Presigned URL to access the Encrypted Object.

Note: Go to next task, But **don't close** the **Browser**.

Step 5: Disable the CMK

20. In the **AWS Management Console**, on the **Services** menu, select **KMS**.

21. Select **Customer managed keys**.

a. Open the **PCI-Data-Key**.

i. Select **Key action**.

ii. Select **Disable**.

a) **Enable** the **Checkmark** against **Confirm that you want to disable this key**.

b) Select **Disable key**.

Note: You can see the **PCI-Data-Key status** as **Disable**.

Step 6: Access the Public Object

17. From the **S3** console, click **Buckets**.

18. Open **secret-123** bucket.

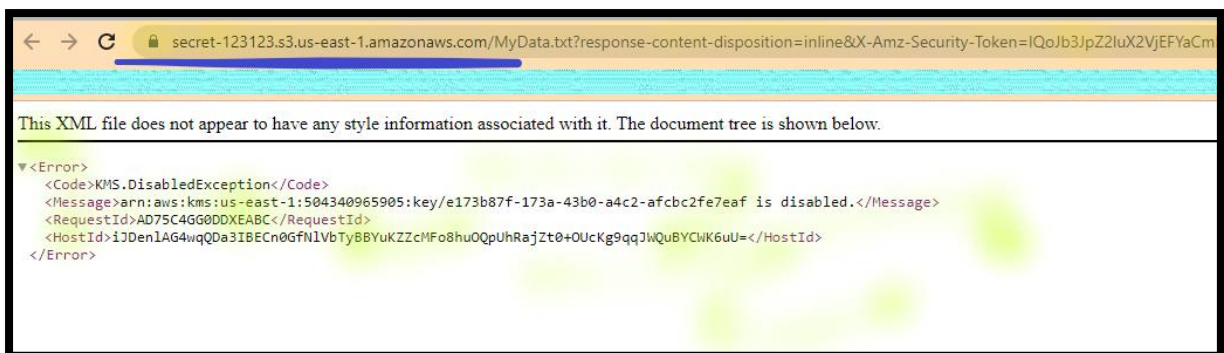
a. Open **mydata.txt**.

b. Click the **Open**.

Note: You are **not able to open the file**.

Note: Notice the error code and message. You are not able to view the file because the key is disabled.

Note: If you are able to view the file, perform a hard refresh (Press CTRL+F5) in your browser as the file may be cached.



Task 5: Create IAM Role

Step 1: Create IAM Policy for AWS S3

51. In the **AWS Management Console**, on the **Services** menu Search and Select **IAM**.

52. Select **Policies**.

53. Click on **Create policy**.

a. Select the **JSON**:

i. **Remove** the **existing policy**.

- ii. **Copy** the policy from **S3ReplicationPolicy.txt**.

Note: **S3ReplicationPolicy.txt** is available with the Lab manual.

- iii. Select **Next: Tags**.
- iv. Select **Next: Review**.
- b. In the **Review Policy** page:
 - i. **Name:** Write **S3ReplicationPolicy**.
 - ii. Click on **Create Policy**.

Note: You will get the message **S3ReplicationPolicy** has been created.

Step 2: Create IAM Roles for AWS S3

54. **From** the **IAM** console:

55. Select **Roles**, click on **Create role**.

- a. In the **Select trusted entity** section:
 - i. **Trusted entity type:** Select **AWS service**.
 - ii. **Common use cases:** Dropdown and select **S3**.
 - a) Select **S3**.
 - iii. Select **Next**.
- b. In the **Add Permissions** section:
 - i. in the **Search box**, type **S3ReplicationPolicy**, press **Enter**.
 - ii. Select **S3ReplicationPolicy**.
 - iii. Select **Next**.
- c. In the **Name, review, and create** section:
 - i. **Role Name:** Write **S3ReplicationRole**.
 - ii. Click **Create role**.

Note: **Wait**, till role **S3ReplicationRole** gets created.

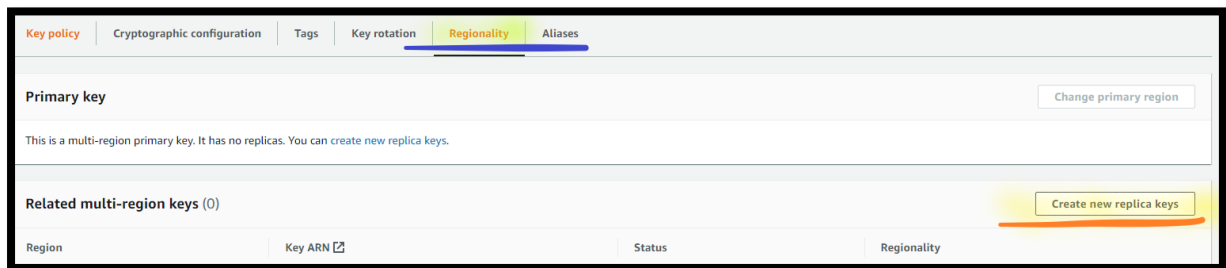
Task 6: AWS S3 Cross-region Replication

Step 1: Create AWS KMS Replica

56. In the **AWS Management Console**, on the **Services** menu, select **KMS**.

57. Select **Customer managed keys**.

- a. Open the **PCI-Data-Key-02**.
 - i. Select the **Regionality**.
 - ii. Select **Create new replica keys**.



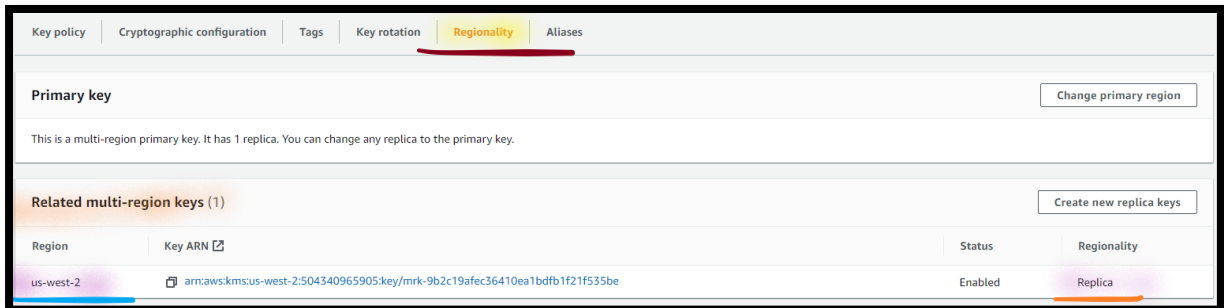
- a) **Choose regions for your new replica key:** Dropdown and select **US West (Oregon)** [us-west-2].
- b) Select **Next**.
- c) Select **Next**.
- d) Select **Next**.
- e) Select **Next**.
- f) Select **Create new replica keys**.
- g) **Enable** the **Checkmark** against **I understand that the values I choose here are not synchronised with any other multi-region key**.

58. **From** the **KMS** console:

- a. Select **Customer managed keys**.
 - i. Open the **PCI-Data-Key-02**.

a) Select the **Regionality**.

Note: You can see the **KMS Replica** details of **us-west-2**.



Step 2: Verify AWS KMS Replica

59. Dropdown and select **US West (Oregon)** region.

60. In the **AWS Management Console**, on the **Services** menu, select **KMS**.

a. Select **Customer managed keys**.

Note: You can see the **PCI-Data-Key-02** details.

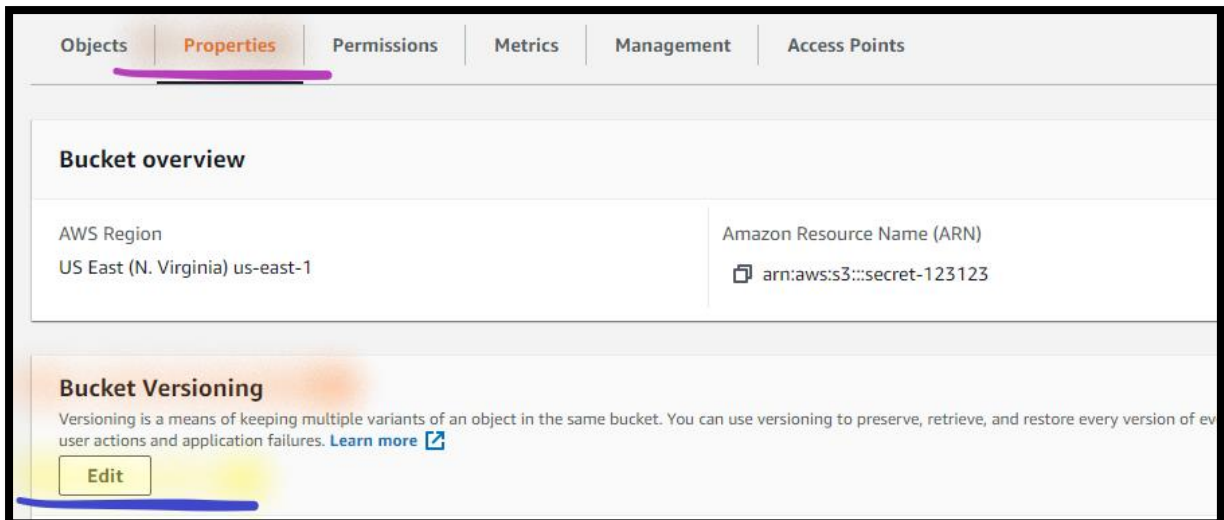
Note: Verify **Key ID** is same in **Primary** and **Replica** region.

Step 3: Enable Versioning for AWS S3 Bucket

61. In the **AWS Management Console**, on the **Services** menu, select **S3**.

62. Select **Buckets**.

- Open **secret-123** bucket.
- Select **properties**.
- In the **Bucket Versioning** section:
 - Select **Edit**.



ii. **Bucket Versioning:** Select **Enable**.



iii. Select **Save Changes**.

Step 4: Create Second AWS S3 Bucket

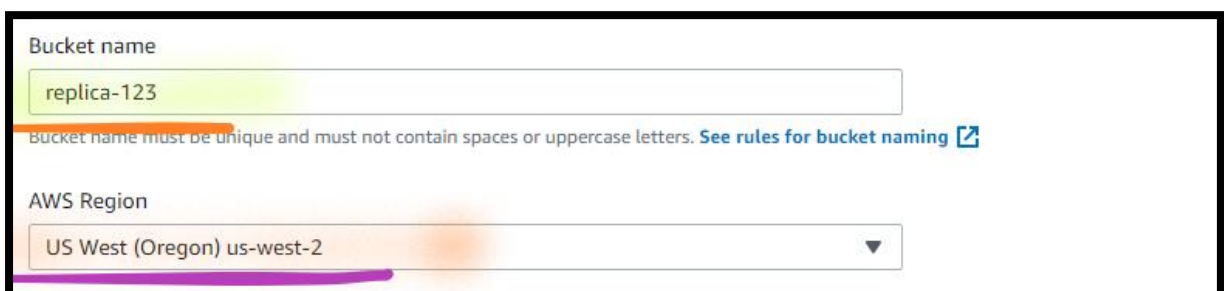
63. From **S3** console:

64. Click **Create bucket**.

a. In the **General configuration** section:

i. **Bucket name:** Write **replica-123**.

ii. **Region:** Dropdown and select **US West (Oregon)**.



- b. In the **Object Ownership** section:
 - i. Select **ACL Enabled**.
 - ii. Select **Bucket owner preferred**.
- c. In the **Block Public Access settings for this bucket** section:
 - i. **Block all public access:** **Deselect** (*remove the checkmark*) the Block all public access option.
 - ii. **Select I acknowledge** that the current settings might result in this bucket and the objects within becoming public.
- d. In the **Bucket Versioning** section:
 - i. **Bucket Versioning:** Select **Enable**.

Note: Leave other details as default.

- e. Click **Create bucket**.

Step 5: Enable Cross-region Replication

65. From **S3** console:

66. Select **Buckets**.

- a. Open **secret-123** bucket.
- b. Select **Management**.
- c. In the **Replication rules** section:
 - i. Select **Create replication rule**:
 - a) In the **Replication rule configuration** section:
 - 1) **Replication rule name:** Write **s3-replication-rule**.

Replication rule name

s3-replication-rule

Up to 255 characters.

Status

Choose whether the rule will be enabled or disabled when created.

☒ Enabled

☐ Disabled

Priority

The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

0

b) In the **Source bucket** section:

- 1) **Choose a rule scope:** Select **Apply to all objects in the bucket.**

Source bucket name

secret-123123

Source Region

US East (N. Virginia) us-east-1

Choose a rule scope

☐ Limit the scope of this rule using one or more filters

☒ Apply to all objects in the bucket

c) In the **Destination** section:

- 1) Select **Choose a bucket in this account.**
- 2) **Bucket name:** Select **Browse:**
 - I. Select the **replica-123** bucket.
 - II. Select **Choose path.**

Destination
You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or see [Amazon S3 pricing](#)

☒ Choose a bucket in this account
☐ Specify a bucket in another account

Bucket name
Choose the bucket that will receive replicated objects.

replica-123 Browse S3

Destination Region
US West (Oregon) us-west-2

III. **Choose a rule scope:** Select **Apply to all objects in the bucket**.

d) In the **IAM role** section:

- 1) Select **Choose from existing IAM roles**.
- 2) **IAM role:** Dropdown and select **S3ReplicationRole**.

IAM role

☒ Choose from existing IAM roles
☐ Enter IAM role ARN

IAM role
S3ReplicationRole View

e) In the **Encryption** section:

- 1) Select **Replicate objects encrypted with AWS KMS**.

☒ **Replicate objects encrypted with AWS KMS**
You can use replication for AWS Key Management Service encrypted objects to replicate data encrypted using AWS KMS across AWS Regions.

- 2) **AWS KMS key for encrypting destination objects:** Select **Choose from your AWS KMS keys**.

3) **AWS KMS key**: Dropdown and select **PCI-Data-Key-02**.

AWS KMS key for encrypting destination objects

☐ AWS managed key (aws/s3)
arn:aws:kms:us-west-2:504340965905:alias/aws/s3

☒ Choose from your AWS KMS keys

☐ Enter AWS KMS key ARN

AWS KMS key

arn:aws:kms:us-west-2:504340965905:key/mrk-e... ▼

[Create key](#)

f) Select **Save**.

Note: You can see the **Replication rules**.

Replication rules (1)

Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. [Learn more](#)

[View details](#) [Edit rule](#) [Delete](#) [Actions](#) [Create replication rule](#)

Replication rule name	Status	Destination bucket	Destination Region	Priority	Scope	Storage class	Replica owner	Replication Time Control	KMS-encrypted objects
s3-replication-rule	Enabled	s3://replica-123	US West (Oregon) us-west-2	0	Entire bucket	Same as source	Same as source	Disabled	Replicate

Step 6: Upload the Object in the Bucket

67. From the **S3** console, click **Buckets**.

68. Open **secret-123** bucket.

- Click **Objects**.
- Click **Upload**.
- Select **Add files**.
- Navigate and select **myreplica.txt** file.

Note: **myreplica.txt** file available with the lab manual.

- Click **Upload**.

Note: **Wait** till all the contents uploaded successfully and can get the message **Upload succeeded**.

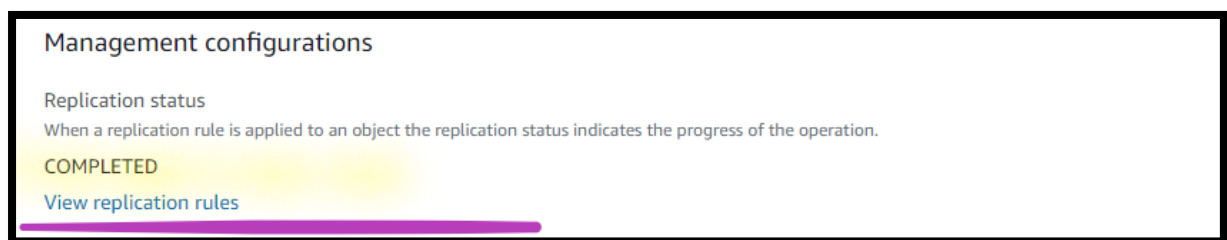
- f. Click **Close**.

Step 7: View the Object status

69. **From** the **Replica-123** bucket console:

- a. Open the **myreplica.txt** file.

Note: **Wait** till the **Replication Status** is **Completed**.

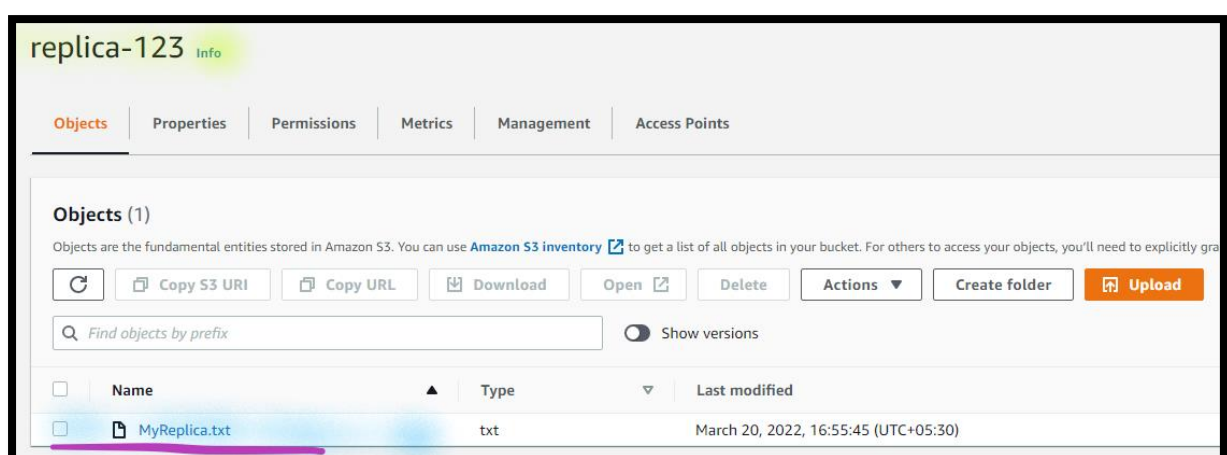


Step 8: Access the Replicated Object

70. **From** the **S3** console, click **Buckets**.

71. Open **replica-123** bucket.

Note: You can see the **myreplica.txt** file.



- a. Open **myreplica.txt** file.
- b. Click the **Open**.

Note: You are **able to open the file**.

Task 4: Cleanup the Environment

Step 1: Terminate EC2 Instances

72. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

73. Click **Instances**.

- a. Select the **Two** instances.
 - i. Click on **Instance state**.
 - ii. Select **Terminate instance**.
 - iii. Select **Terminate**.

Step 2: Delete AMIs

74. **From** the **EC2** console, click **AMIs**.

- a. Select **WindowsServerImage-01**.
 - i. Select **Actions**.
 - ii. Select **Deregister AMI**.
 - iii. Select **Deregister AMI**.
- b. Select **WindowsServerImage-02**.
 - i. Select **Actions**.
 - ii. Select **Deregister AMI**.
 - iii. Select **Deregister AMI**.

Step 3: Delete Snapshots

75. **From** the **EC2** console, click **Snapshots**.

- a. Select **All** the snapshots.

- i. Select **Actions**.
- ii. Select **Delete Snapshot**.
 - a) In the **prompt**, type **delete**.
 - b) Select **Delete**.

Step 4: Delete Volumes

76. **From** the **EC2** console, click **Volumes**.
- a. Select **All** the volumes.
 - i. Select **Actions**.
 - ii. Select **Delete volume**.
 - a) In the **prompt**, type **delete**.
 - b) Select **Delete**.

Step 5: Delete the S3 Bucket

77. In the **AWS Management Console**, on the **Services** menu, click **S3**.
78. Select **Buckets**.

Delete the First bucket

79. Select **secret-123** bucket.
- a. Select **Empty**.
 - b. **Type** **permanently delete** to **Delete all the objects**.
 - c. Select **Empty**.
 - d. Select **Exit**.
80. Select **secret-123** bucket.
- a. Select **Delete**.
 - b. **Type** **secret-123** bucket name, to **Delete bucket**.
 - c. Select **Delete bucket**.

Delete the Second bucket

81. Select **replica-123** bucket.
- a. Select **Empty**.
 - b. Type **permanently delete** to **Delete all the objects**.
 - c. Select **Empty**.
 - d. Select **Exit**.
82. Select **replica-123** bucket.
- a. Select **Delete**.
 - b. Type **replica-123** bucket name, to **Delete bucket**.
 - c. Select **Delete bucket**.

Step 6: Schedule the CMK for deletion

83. In the **AWS Management Console**, on the **Services** menu Search and Select **KMS**.
84. Dropdown and select **US East (N. Virginia)** region.
85. Select **Customer managed keys**.

Delete the First CMK

- a. Open the **PCI-Data-Key**.
 - i. Select **Key actions**.
 - ii. Select **Schedule key deletion**.
 - iii. **Waiting period (days)**: Type **7**.
 - iv. **Select** **Confirm that you want to schedule these keys for deletion after a 7 day waiting period**.
 - v. Select **Schedule deletion**.

Delete the Second CMK

- b. Open the **PCI-Data-Key-02**.
 - i. Select **Key actions**.
 - ii. Select **Schedule key deletion**.

- iii. **Waiting period (days):** Type **7**.
- iv. **Select** *Confirm that you want to schedule these keys for deletion after a 7 day waiting period.*
- v. Select **Schedule deletion**.

Delete the Replica CMK

86. **From** the **KMS** console:

87. Dropdown and select **US West (Oregon)** region.

- a. Open the **PCI-Data-Key-02**.
 - i. Select **Key actions**.
 - ii. Select **Schedule key deletion**.
 - iii. **Waiting period (days):** Type **7**.
 - iv. **Select** *Confirm that you want to schedule these keys for deletion after a 7 day waiting period.*